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An expert-level generalist AI for abdominal CT diagnosis

Qi Zhang†, Jianpeng Zhang†, Weiwei Cao†, Zilin Lu†, Wanxing Chang†, Haonan Ding† *et al.*

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INTROD'UCTION: Radiology is critical for modern clinical diagnosis, enabling the detection and characterization of diverse diseases. A long-standing vision in the field is to realize a generalist artificial intelligence (AI) capable of expert-level diagnostic support for varied real-world scenarios. Contrast-enhanced abdominal computed tomography (CT) poses one of the most complex challenges, requiring assessment of dozens of organs and hundreds of conditions. Despite milestones in AI for specific domains, current AI solutions remain limited in disease coverage and accuracy. They largely depend on supervised learning with expensive manual annotation, restricting scalability and applicability in open-ended clinical contexts.

RATIONALE: Vision-language learning offers an alternative to annotation-intensive supervised learning by training models on paired image-text data, with the core principle of pulling visual and textual embeddings from the same patient closer in a shared representation space while separating those from different patients. However, methods successful in natural image domains do not readily transfer to abdominal CT because of its anatomical complexity and sparse diagnostic signals. To address these barriers, we developed RADAR (Rapid Abdominal Diagnosis with AI and Radiology), a fine-grained vision-language framework that decomposes CT volumes into anatomical units; aligns them with diagnostic textual reports via adaptive contrastive learning; and achieves scalable, versatile, and interpretable diagnosis without additional manual labeling.

RESULTS: We constructed RAD-CT, a large abdominal CT dataset, consisting of 424,911 examinations and 15 million anatomy-wise pairs for fine-grained, large-scale training. Evaluated across 18 anatomical structures and 146 imaging findings, RADAR achieved an area under the receiver operating characteristic curve (AUC) of 0.913 (95% confidence interval, 0.911 to 0.915) for the diagnosis of 146 findings in a real-world internal cohort of 39,160 examinations and AUCs of 0.874 to 0.912 across eight external centers. In pathology-confirmed evaluation for four types of cancers (liver, pancreas, stomach, and colorectum), performance reached AUCs of 0.891 to 0.984. In another cross-population cohort, RADAR achieved an AUC of 0.883 without any fine-tuning. In acute abdominal conditions, which had been excluded from initial training, RADAR maintained high performance with an AUC of 0.904, underscoring strong generalization to urgent, unfamiliar scenarios. In a reader study with 26 radiologists from 14 centers, RADAR outperformed most participants and increased sensitivity by ~10% in collaborative use. RADAR also provided attention maps to highlight diagnostic-relevant visual cues, thereby facilitating better interpretation, a feature of considerable value in clinical practice.

CONCLUSION: RADAR is a generalist AI model built on a scalable vision-language framework trained directly on large-scale clinical reports without manual annotation. It achieved broad diagnostic coverage and high accuracy in abdominal CT interpretation and improved radiologists' sensitivity when used as an assistive reading tool. This work suggests that large-scale vision-language learning may provide a practical path toward efficiently developing generalist AI systems for complex and diverse radiologic interpretation tasks. □

Corresponding authors: Tingbo Liang (liangtingbo@zju.edu.cn); Ling Zhang (ling.z@alibaba-inc.com); Qi Zhang (qi.zhang@zju.edu.cn); Jianpeng Zhang (jianpeng.zhang0@gmail.com); Wenbo Xiao (xiaowenbo@zju.edu.cn) †These authors contributed equally to this work. Cite this article as Q. Zhang *et al.*, *Science* 393, eaec6129 (2026). DOI: 10.1126/science.aec6129

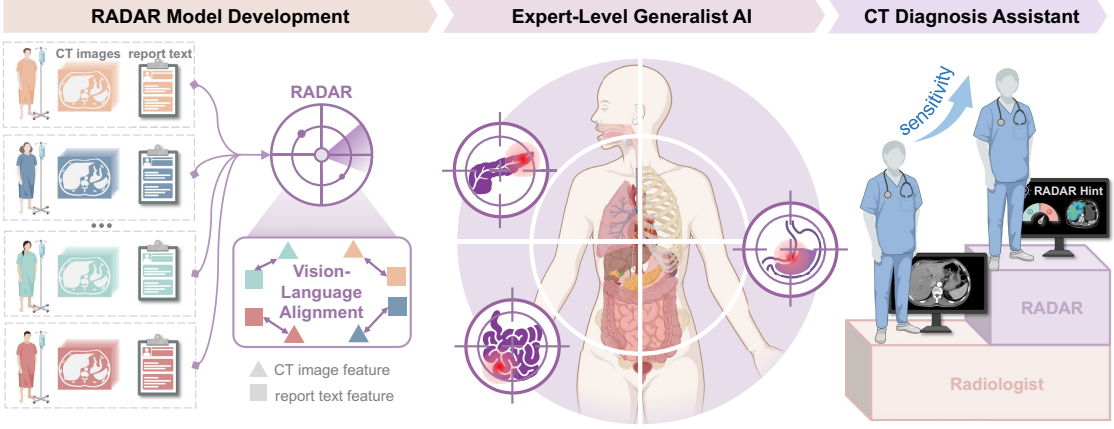


Illustration of the RADAR model development and deployment. Applying vision-language learning to a large, unannotated dataset of abdominal CT images and reports, RADAR provides expert-level diagnosis across a wide range of anatomies and findings. It serves as a generalist AI assistant, empowering radiologists and enhancing their diagnostic performance. [Figure created with BioRender.com]

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IMAGING

An expert-level generalist AI for abdominal CT diagnosis

Qi Zhang[1,2,3]†*, Jianpeng Zhang[4,5,6]†*, Weiwei Cao[4,5,6]†, Zilin Lu[4,7]†, Wanxing Chang[4,5]†, Haonan Ding[1]†, Cao Chen[1], Zhi Li[8], Xing Xue[8], Sinuo Wang[9], Shaoteng Zhang[4,7], Yutong Xie[10], Yong Xia[7], Qi Wu[9], Zhongyi Shui[6], Xi Li[6], Zhilin Zheng[4,5], Yanjie Zhou[4,5,6], Tony C.W. Mok[4,5], Yingda Xia[11], Hongkan Wang[1], Xianghua Ye[12], Tao Ma[1,13], Jie Peng[14], Xiaoguang Wang[15], Jian Ding[16], Yuming Gao[17], Huazhen Ye[18], Yiping Liu[19], Dongjie Chen[20], Zhaomin Ni[21], Jianwen Ning[22], Wei Zhang[1,23], Jian Liu[24], Chaohui Yu[25], Shenghong Ju[26], Jianfeng Zhang[4,5], Wenbo Xiao[8]*, Ling Zhang[11]*, Tingbo Liang[1,2,3]*

Artificial intelligence (AI) in radiology aspires to deliver expert-level diagnosis across diverse clinical tasks, yet existing supervised strategies remain limited in scope. We developed RADAR, a generalist vision-language model trained on more than 400,000 contrast-enhanced abdominal computed tomography (CT) examinations and 15 million anatomy-wise image-text pairs, learning directly from clinical reports without manual annotation. Throughout internal and external evaluations across multiple centers and varied clinical scenarios, RADAR achieved high diagnostic performance and robust generalization for 18 anatomical structures and 146 imaging findings. In a reader study, RADAR assistance increased the diagnostic sensitivity of 26 radiologists by ~10%. RADAR offers a scalable, versatile, and interpretable solution for abdominal CT, demonstrating that generalist AI can match human experts in general and complicated radiology tasks.

The ultimate ambition of radiology is to realize an expert-level generalist artificial intelligence (AI) capable of delivering accurate and comprehensive diagnostic support across diverse real-world clinical settings (*1*, *2*). Although AI has achieved landmark breakthroughs in multiple domains (*3*–*11*), no viable solution has yet emerged for achieving this goal in radiology. Generalist AI in radiology remains at an early stage, with current solutions often exhibiting limited coverage of disease types and suboptimal performance (*12*–*14*). Contrast-enhanced abdominal computed tomography (CT) represents one of the most demanding challenges in this field, involving the assessment of dozens of organs and the identification of hundreds of potential diseases, as well as requiring years of specialized expertise for accurate interpretation. This underscores the urgent need for technological evolution that transcends the constraints of limited functionalities and delivers a generalist AI diagnosis.

Current AI techniques for medical imaging largely follow a supervised learning paradigm, in which abundant, high-quality annotations have enabled specialist AI to achieve, and sometimes surpass, expert performance in single-disease diagnostic settings (*15*–*20*). However, such approaches are constrained by costly manual annotation, exhibit poor scalability, and remain limited in open-ended clinical scenarios. The emerging paradigm of vision-language learning offers a promising alternative, enabling models to learn directly from associations between image and text descriptions rather than relying on manual labels (*21*–*26*). This approach has delivered remarkable results in the general vision domain (*27*) and has been explored in CT imaging by early efforts including Merlin (*12*), CT-CLIP (*13*), and BIUD (*28*). However, particular challenges exist in abdominal CT, in which contextual information is highly complex and diagnostic signals are extremely sparse relative to the whole three-dimensional (3D) volume, posing substantial difficulties for the direct translation of advances from the natural to the medical domain. We previously developed fVLM (*14*), a fine-grained vision-language framework capable of recognizing 54 disease categories in abdominal CT, providing a proof of concept for this approach. However, both diagnostic accuracy and disease coverage remain to be substantially improved, motivating the development of more-advanced generalist AI for radiology.

In this study, we present Rapid Abdominal Diagnosis with AI and Radiology (RADAR), a vision-language generalist model for broad diagnosis of contrast-enhanced abdominal CT. RADAR decomposes CT volumes into anatomical units and aligns each with corresponding diagnostic report descriptions through contrastive learning, bringing visual and textual features from the same individual closer while separating those from different individuals at the anatomy level. To enable this alignment, an anatomical perception module localizes abdominal structures and encodes them into anatomy-level representations, and an adaptive contrastive strategy leverages semantic similarities across patients to strengthen cross-modal associations. To overcome data scarcity, we created a fine-grained abdominal CT dataset comprising 424,911 examinations and 1.5 million volume-wise image-text pairs, yielding more than 15 million anatomy-wise image-text pairs for large-scale training. To comprehensively assess RADAR, we established a benchmark covering 18 anatomical structures and 146 findings, evaluated its performance on large real-world and multiple external cohorts (including cross-population, acute abdominal, and pathology-confirmed cases), and conducted a reader study to demonstrate its clinical utility in assisting radiologists.

Results

Development of RADAR

The architecture of RADAR comprises three components (Fig. 1A): a visual encoder that extracts high-dimensional feature representations from CT images, a mask decoder that produces voxel-level anatomical segmentation maps to derive structure-specific visual features, and a text encoder that converts corresponding anatomical subreports

[1]Department of Hepatobiliary and Pancreatic Surgery, The First Affiliated Hospital, Zhejiang University School of Medicine, Hangzhou, Zhejiang, China. [2]Zhejiang Provincial Key Laboratory of Pancreatic Disease, The First Affiliated Hospital, Zhejiang University School of Medicine, Hangzhou, Zhejiang, China. [3]MOE Joint International Research Laboratory of Pancreatic Diseases, The First Affiliated Hospital, Zhejiang University School of Medicine, Hangzhou, Zhejiang, China. [4]Alibaba DAMO Academy, Hangzhou, Zhejiang, China. [5]Hupan Laboratory, Hangzhou, Zhejiang, China. [6]College of Computer Science and Technology, Zhejiang University, Hangzhou, Zhejiang, China. [7]Department of Radiology, Ningbo No. 2 Hospital, Ningbo, Zhejiang, China. [8]Department of Radiology, The First Affiliated Hospital, Zhejiang University School of Medicine, Hangzhou, Zhejiang, China. [9]Australian Institute for Machine Learning, Adelaide University, Adelaide SA, Australia. [10]Department of Computer Vision, Mohamed bin Zayed University of Artificial Intelligence, Masdar City, Abu Dhabi, United Arab Emirates. [11]Alibaba DAMO Academy, Washington, DC, USA. [12]Department of Radiation Oncology, The First Affiliated Hospital, Zhejiang University School of Medicine, Hangzhou, Zhejiang, China. [13]Department of General Surgery, The First Division Hospital of Xinjiang Production and Construction Corps, Akesu, Xinjiang, China. [14]Department of Radiology, The First Division Hospital of Xinjiang Production and Construction Corps, Akesu, Xinjiang, China. [15]Department of Surgery, Affiliated Hospital of Jiaxing University, Jiaxing, Zhejiang, China. [16]Department of Radiology, Affiliated Hospital of Jiaxing University, Jiaxing, Zhejiang, China. [17]Department of General Surgery, Jixi County People's Hospital, Xuancheng, Anhui, China. [18]Department of Radiology, People's Hospital of Jingning She Autonomous County, Lishui, Zhejiang, China. [19]Department of Radiology, Shengzhou People's Hospital, Shaoxing, Zhejiang, China. [20]Department of General Surgery, Haining People's Hospital, Jiaxing, Zhejiang, China. [21]Department of Radiology, Anji County People's Hospital, Huzhou, Zhejiang, China. [22]Department of Emergency, The First Affiliated Hospital, Zhejiang University School of Medicine, Hangzhou, Zhejiang, China. [23]Department of General Surgery, The First People's Hospital of Yuhang District, Hangzhou, Zhejiang, China. [24]Department of Surgical Oncology, The People's Hospital of Beilun District, Ningbo, Zhejiang, China. [25]Department of Gastroenterology, The First Affiliated Hospital, Zhejiang University School of Medicine, Hangzhou, Zhejiang, China. [26]Nurturing Center of Jiangsu Province for State Laboratory of AI Imaging & Interventional Radiology (Southeast University); Department of Radiology, Zhongda Hospital, Medical School of Southeast University, Nanjing, Jiangsu, China. ***Corresponding author. Email: liangtingbo@zju.edu.cn (T.L.); ling.z@alibaba-inc.com (L.Z.); qi.zhang@zju.edu.cn (Q.Z.); jianpeng.zhang0@gmail.com (Jianp.Z.); xiaowenbo@zju.edu.cn (W.X.)** †These authors contributed equally to this work.

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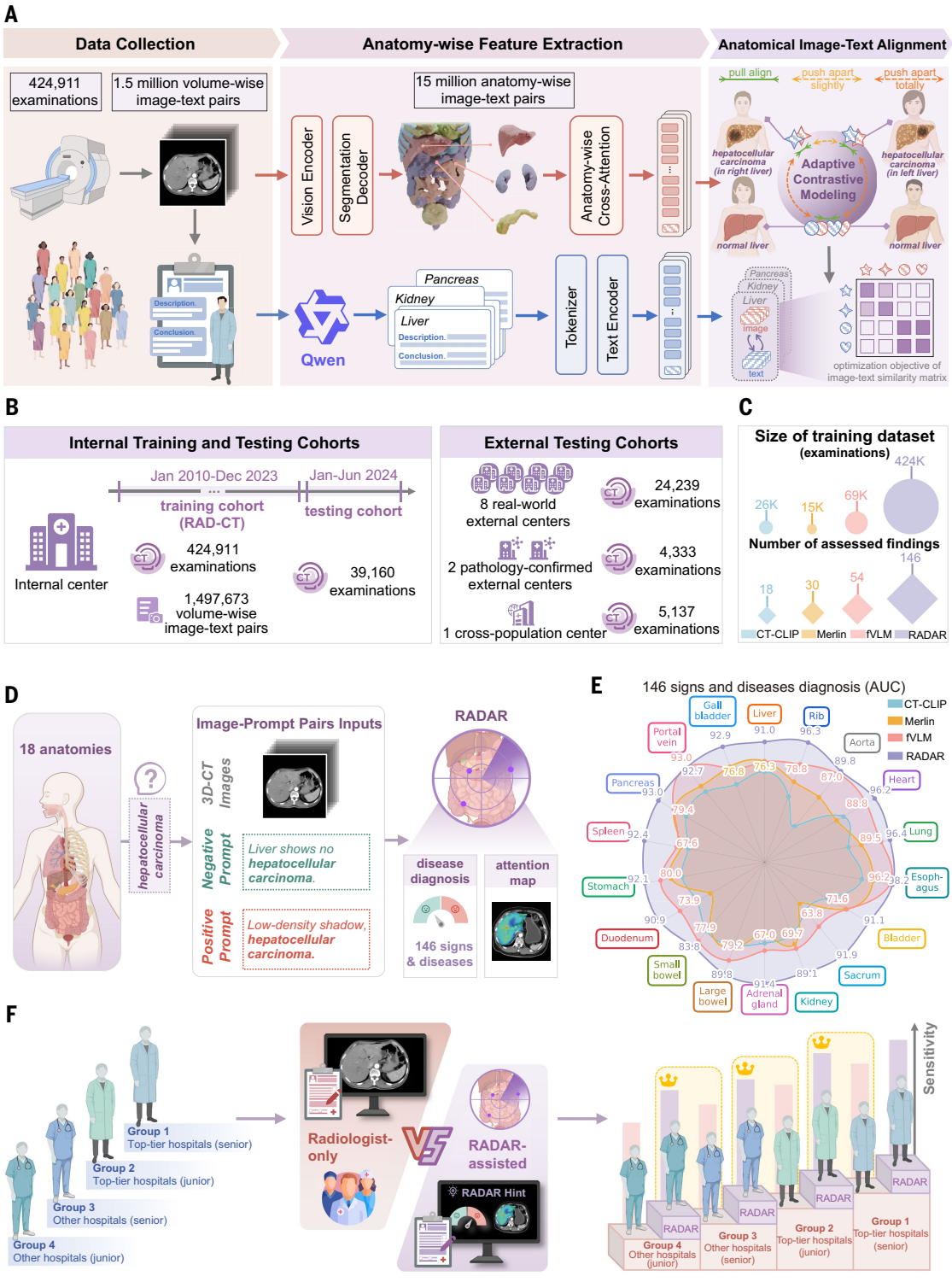


Fig. 1. A schematic overview of the development, evaluation, and clinical application of RADAR. (**A**) RADAR was trained on a large-scale abdominal CT image and report dataset comprising 424,911 CT examinations and 1.5 million CT volume-wise image-text pairs. We applied a fine-grained learning framework to align anatomy-wise vision-language features, using an anatomical perception module to automatically localize specific anatomical structures. Then anatomy-wise image and text features were aligned under the guidance of the adaptive contrastive modeling module. (**B**) The training cohort (RAD-CT) was constructed from a dataset spanning over 10 years from a single center. A separate 6-month dataset from the same center served as an internal testing cohort, and data from eight real-world centers, two pathology-confirmed external centers, and one cross-population center were collected for extensive external validation. (**C**) Data scale and diagnostic capability: RADAR versus other methods. (**D**) Example of the RADAR inference workflow, in which diagnosis is performed through prompt-based querying. (**E**) RADAR achieves an average 0.913 AUC across 18 anatomical structures spanning 146 signs and diseases in the internal cohort, outperforming existing state-of-the-art vision-language models CT-CLIP, Merlin, and fVLM. (**F**) RADAR serves as a generalist AI assistant, integrating into clinical workflows to improve diagnostic performance for radiologists of all experience levels and in all practice settings. [Figure created with BioRender.com]

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into textual features. The model was trained on ~15 million anatomy-wise image-text pairs. These pairs were derived from the Rapid Abdominal Diagnosis–CT (RAD-CT) cohort by anatomically decomposing volume-wise image-text data from more than 400,000 cases at a single high-volume center (Fig. 1, B and C). Once trained, the joint feature space enables prompting-based diagnosis without any fine-tuning. For every diagnostic query, two textual prompts are designed: a positive prompt describing its presence (for example, "colitis") and a negative prompt describing its absence (for example, "no evidence of colitis") (Fig. 1D). Both prompts are encoded by the text encoder and compared with the corresponding anatomy-specific image feature using cosine similarity. The prompt associated with the higher similarity score is assigned as the prediction. This prompting strategy allows RADAR to handle a broad and variable range of diagnostic tasks without task-specific retraining (Fig. 1E), facilitating deployment in clinical settings where the spectrum of potential findings is not fixed in advance. Furthermore, RADAR provides interpretability through attention maps that highlight image regions most influential to each prediction, offering clinicians transparent visual cues to support diagnostic reasoning and build trust in AI-assisted decision-making (Fig. 1F).

Internal real-world evaluation

To assess the performance of RADAR, we first evaluated it on a large-scale, internal real-world testing cohort of 39,160 consecutive examinations acquired during an independent period, benchmarking against three representative vision-language models (VLMs) for CT diagnosis (CT-CLIP, Merlin, and fVLM), each pretrained on RAD-CT for this study. Across a total of 146 clinically relevant findings (see Materials and methods "Benchmark and evaluation" and table S1), RADAR achieved a mean area under the receiver operating characteristic curve (AUC) of 0.913 [95% confidence interval (CI), 0.911 to 0.915], exceeding the second-best model fVLM (0.776; 95% CI, 0.773 to 0.779) by 0.137 ($P < 0.001$), and substantially surpassing the other two models (table S2).

Grouping findings by 18 anatomical structures revealed that RADAR maintained high and consistent performance, with AUCs ranging from 0.838 to 0.982, outperforming the three comparison models in almost all structures (Fig. 2A). This stability extended across various functional systems, including respiratory, musculoskeletal, urinary, digestive, endocrine, hematologic, and circulatory systems, where AUCs ranged from 0.896 to 0.964 (Fig. 2B and table S3). By anatomical type, RADAR demonstrated high performance for both solid organs (AUC mean, 0.918; 95% CI, 0.915 to 0.920) and hollow organs (AUC mean, 0.907; 95% CI, 0.904 to 0.911) (Fig. 2C and table S4). Consistently, RADAR achieved similar performance for disease-related (AUC mean, 0.914, 95% CI, 0.911 to 0.918) and sign-related findings (AUC mean, 0.911, 95% CI, 0.909 to 0.913) (Fig. 2D and table S5). In addition, RADAR exhibited robust performance for both benign and malignant findings, achieving mean AUCs of 0.910

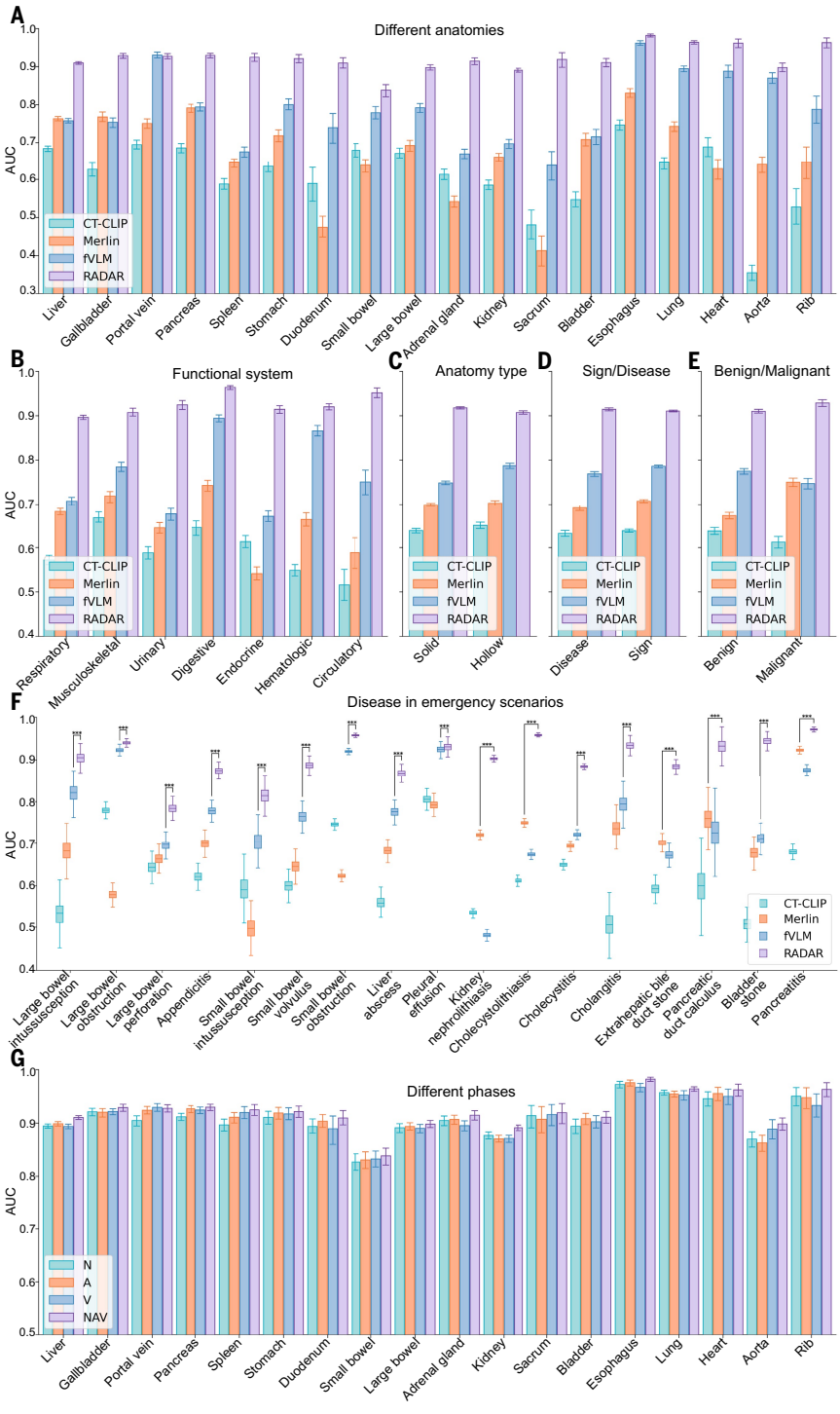


Fig. 2. Internal real-world evaluation of RADAR. (**A**) Comparison of diagnostic performance (AUC) across 18 anatomical structures, evaluated on 146 findings in 39,160 examinations. RADAR outperformed three representative vision-language CT models: CT-CLIP, Merlin, and fVLM. (**B** to **E**) Subgroup analyses showing RADAR's performance across (B) seven functional systems (respiratory, musculoskeletal, urinary, digestive, endocrine, hematologic, and circulatory), (C) two anatomical types (solid organs and hollow organs), (D) findings categorized as imaging signs versus diseases, and (E) diseases stratified by benign versus malignant categories. (**F**) Performance comparison of different methods on 17 common acute abdominal conditions in emergency settings, with none of the models trained on emergency data. Box plots depict the distribution of AUCs derived from 1000 bootstrap resamples. (**G**) Performance across different imaging phases. N, nonenhanced phase; A, arterial phase; V, venous phase; NAV, ensemble of all three phases, in which the maximum prediction score among the three phases was used as the ensemble prediction score. Error bars, 95% CIs. *** $P < 0.001$ with DeLong test.

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(95% CI, 0.906 to 0.914) and 0.929 (95% CI, 0.921 to 0.936), respectively (Fig. 2E and table S6). Although RADAR was trained with anatomy-level alignment, it also maintained strong performance in recognizing multiorgan pathologies, such as portal vein hypertension (PVH)–related comorbidities that include concurrent cirrhosis, PVH, esophagus varices, gastric fundal varices, and splenomegaly (fig. S1).

Despite not being trained on emergency data, RADAR also exhibited strong performance in emergency scenarios. We evaluated it on a dataset comprising 27,267 emergency cases, including 17 common acute abdominal conditions selected on the basis of their prevalence (Materials and methods "Benchmark and evaluation" and table S1). RADAR achieved a mean AUC of 0.904 (95% CI, 0.900 to 0.909) (Fig. 2F), despite high variability and substantial clinical complexity of emergency settings (*29*, *30*). RADAR demonstrated stable performance across different imaging phases, with multiphase fusion achieving the best performance (Fig. 2G). Furthermore, we observed that RADAR's performance improved consistently with increasing amounts of training data, highlighting its strong scalability and potential for further enhancement with more data (fig. S2).

We then compared RADAR with the advanced supervised learning models, Vision Transformer (ViT) (*31*) and nnU-Net+ (*32*) (Materials and methods "Details of compared methods"), in the internal testing cohort. Across the 146 findings, RADAR significantly outperformed ViT (AUC mean, 0.787, 95% CI, 0.784 to 0.791) and nnU-Net+ (AUC mean, 0.868, 95% CI, 0.865 to 0.870) by 0.126 and 0.045 (both $P < 0.001$), respectively (fig. S3A). Notably, RADAR maintained strong performance for both low- and high-frequency findings, whereas both supervised methods performed markedly worse on low-frequency findings (fig. S3B).

External real-world evaluation

We retrospectively collected the consecutive testing cohorts from eight external centers to evaluate the generalizability and robustness of RADAR in real-world, multicenter clinical scenarios, encompassing a broad spectrum of patient demographics and heterogeneous imaging acquisition protocols. These examinations were obtained from diverse clinical settings, including emergency, inpatient, outpatient, and preventive health checkup. A total of 24,239 contrast-enhanced abdominal CT examinations were included, covering whole-abdomen, upper-abdomen, mid-abdomen, lower-abdomen, and pelvic scans. The number of annotated findings per center ranged from 68 to 136, with labels extracted from CT reports (Materials and methods "Benchmark and evaluation" and table S1). Across the eight external centers, RADAR achieved an average AUC of 0.895 (95% CI, 0.889 to 0.900), substantially outperforming the second-best model, fVLM, by 0.176 ($P < 0.001$). Notably, RADAR consistently outperformed all three baseline methods at every individual center, with single-center AUCs ranging from 0.874 to 0.912, demonstrating consistently high-level performance across institutions with varying operational conditions and imaging protocols (Fig. 3A and table S7).

Further analysis revealed that RADAR maintained stable generalization performance across 18 anatomical structures when evaluated on internal versus external centers (Fig. 3B). Notably, RADAR's performance was comparable between internal and external centers for more than half of the structures (for example stomach, liver, and sacrum), with AUC differences below 0.01. For some other structures (for example duodenum, small bowel, and spleen), we observed well-generalized performance across external centers. These results further demonstrate RADAR's generalizability and robustness across heterogeneous real-world clinical settings and diverse anatomical contexts.

Compared with the supervised methods, RADAR's advantage was more evident in the external cohorts. Compared with the internal cohort, AUCs decreased by 0.018 for RADAR, 0.032 for ViT, and 0.050 for nnU-Net+ (fig. S3C). The gap was particularly clear for low-frequency findings, in which the supervised learning models' performance declined more sharply (fig. S3D).

Generalization to pathology-confirmed testing cohorts

A key question is whether RADAR that trained with radiology reports maintains its performance when tested against the gold-standard ground truth of pathology-confirmed diagnoses. To answer this, we collected two additional independent external cohorts, each comprising cases from four types of cancer confirmed by either surgical or biopsy pathology. In total, the two cohorts had 4333 cases (648 with colorectal cancer, 388 with pancreatic cancer, 537 with gastric cancer, 383 with hepatocellular carcinoma, and 2377 normal controls). As expected, RADAR achieved consistently high performance across the four cancer types (Fig. 3C). In pathology center 1, AUCs ranged from 0.922 to 0.984, and in the other center, AUCs ranged from 0.891 to 0.972. Notably, in the external real-world testing cohorts from the eight centers, RADAR's AUCs were 0.905 to 0.987 for colorectal cancer, 0.909 to 0.989 for pancreatic cancer, 0.899 to 0.987 for gastric cancer, and 0.863 to 0.974 for hepatocellular carcinoma, which were comparable to performances observed in the pathology-confirmed testing cohorts. These results suggest that, although RADAR initially learned from radiology reports, its diagnostic capability generalizes well to cohorts with pathology-confirmed labels, underscoring its broad generalizability and potential clinical utility.

Generalization to a cross-population cohort

To rigorously evaluate the generalizability of RADAR in a population with a distinct geographic and ethnic background from the Asian cohorts, we assessed it on the public Merlin abdominal CT test set (Merlin-CT-Test), which contains 5137 CT scan images from Stanford Hospital in the United States. This dataset predominantly consists of Western patients. Following the Merlin benchmark, the evaluation covered 30 imaging findings.

We evaluated two distinct settings. First, we performed direct testing of different models (Materials and methods "Details of compared methods"), including BiomedCLIP (*33*), MedGemma (*34*), Lingshu (*35*), and RADAR, that had not been trained on Merlin data, constituting a highly challenging cross-population generalization test. Second, we compared models under conditions in which training and fine-tuning on Merlin dataset (Merlin-CT-Train) was allowed, including Merlin, RADAR (trained on Merlin-CT-Train from scratch) and RADAR+. Here, RADAR+ refers to a version of RADAR fine-tuned on the Merlin-CT-Train dataset (Materials and methods "RADAR+: Extending diagnostic scopes"). The evaluation scope included two subsets: (i) in-scope findings, comprising 21 findings from the total 30 findings for which RADAR can perform anatomy-level diagnosis; 9 out-of-scope findings were excluded from this subset owing to either the lack of a clear organ association or RADAR's current absence of organ segmentation capability; and (ii) all findings, in which all 30 findings were evaluated, enabled by RADAR's expanded capability after fine-tuning.

RADAR demonstrated outstanding cross-population generalization, achieving an AUC of 0.883 (95% CI, 0.871 to 0.891) on the 21 in-scope findings subset, which represents a clear advantage over other vision-language models (Table 1). RADAR's external evaluation performance was also significantly higher than Merlin's internal performance (+0.071, $P < 0.05$). When trained from scratch on Merlin-CT-Train, RADAR achieved an AUC of 0.888 (95% CI, 0.879 to 0.897) on the 21 in-scope findings, outperforming Merlin trained on the same dataset. Moreover, the RADAR+ fine-tuning strategy yielded a significant improvement compared with RADAR (+0.030, $P < 0.05$) and effectively expanded diagnostic coverage to all 30 findings. A detailed performance comparison of different models on each of the 30 findings is provided in table S8.

Reader studies

To evaluate RADAR's clinical utility, we conducted a two-session reader study (Materials and methods "Reader studies") comparing its performance with that of radiologists of varying experience and quantifying

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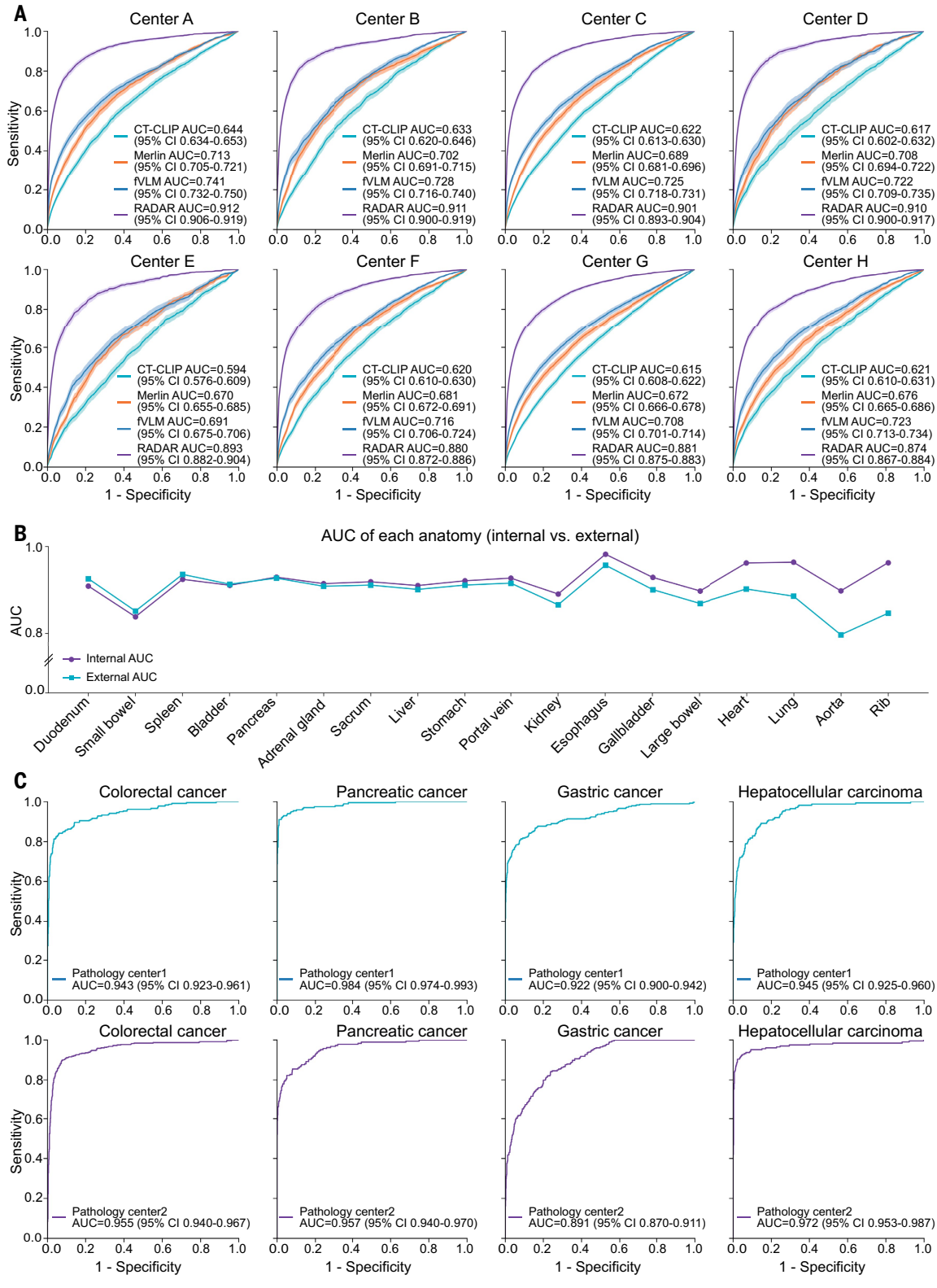


Fig. 3. External evaluation. (**A**) Performance of RADAR and three compared methods (CT-CLIP, Merlin, and fVLM) on external real-world testing cohorts from eight centers. The colored bands indicate the standard error of the ROC curves, with each band's central line representing the mean AUC calculated across multiple findings, thereby illustrating the overall multicenter performance of each method. (**B**) Comparison of RADAR's AUC between the internal real-world testing cohort and external multicenter real-world testing cohorts across 18 anatomical structures. For most anatomical structures, RADAR achieved external multicenter AUC values comparable to those from internal data, highlighting its strong generalizability and robustness across institutions and heterogeneous imaging acquisition settings. Data are arranged in ascending order of internal-external AUC differences. (**C**) Performance of RADAR at two external centers with pathology-confirmed labels. Four cancer diagnostic tasks were evaluated, including colorectal cancer, pancreatic cancer, gastric cancer, and hepatocellular carcinoma.

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Table 1. Cross-population evaluation on the public Merlin-CT-Test benchmark. The upper section compares the AUC performance of RADAR with other vision-language models, including BiomedCLIP, MedGemma, and Lingshu, on the unseen Merlin-CT data, highlighting their zero-shot generalization capabilities. The lower section presents AUC results for methods trained directly with Merlin-CT-Train data, specifically Merlin, RADAR trained from scratch, and RADAR+, where "RADAR+" denotes a fine-tuned version of RADAR on the Merlin-CT dataset.

Methods	Train dataset	Anatomy-specific findings (in-scope, $n = 21$) (95% CI)	All findings ($n = 30$) (95% CI)
Evaluation of models without training on Merlin-CT-Train dataset			
BiomedCLIP	Medical multimodal datasets	0.574 (0.563 to 0.586)	0.602 (0.592 to 0.612)
MedGemma1.5	Medical multimodal datasets	0.555 (0.547 to 0.563)	0.558 (0.551 to 0.566)
Lingshu	Medical multimodal datasets	0.581 (0.570 to 0.592)	0.608 (0.598 to 0.617)
RADAR	RAD-CT	0.883 (0.871 to 0.891)	/
Evaluation of models trained on Merlin-CT-Train dataset			
Merlin	Merlin-CT-Train	0.812 (0.801 to 0.824)	0.816 (0.806 to 0.826)
RADAR	Merlin-CT-Train	0.888 (0.879 to 0.897)	/
RADAR+	(pretrain) RAD-CT + (fine-tune) Merlin-CT-Train	0.918 (0.911 to 0.925)	0.876 (0.868 to 0.884)

its assistance in their reporting. The study included contrast-enhanced abdominal CT scans from 300 patients, covering 61 distinct imaging findings. The reader cohort comprised 26 radiologists from 14 institutions (table S9), stratified into four groups by the radiologist's experience and institutional tier: senior radiologists from top-tier hospitals (G1, $n = 4$), junior radiologists from top-tier hospitals (G2, $n = 6$), senior radiologists from other hospitals (G3, $n = 7$), and junior radiologists from other hospitals (G4, $n = 9$).

In the unassisted session, radiologists independently interpreted these CT scans and wrote diagnostic reports. Although their specificity was uniformly high (98.8%, 95% CI, 98.6 to 98.9%), the sensitivity varied substantially, correlating with their professional experience and institutional tier. The operating points of most radiologists lay beneath RADAR's curve, with only three senior radiologists achieving performance slightly better than that of RADAR (Fig. 4A). After a mandatory washout period of at least 1 month, the same radiologists reinterpreted these CT scans, which were presented in a randomized patient order for the RADAR-assisted session. RADAR provided a set of predicted findings together with attention maps highlighting suspicious regions visualized using the Grad-CAM technique (*36*) (figs. S4 and S5). Notably, RADAR has the ability to highlight a wide range of findings, including diverse anatomical structures, functional systems, organ types, signs, and both benign and malignant disease findings (fig. S6).

The assistance of RADAR yielded substantial improvements in these radiologists' diagnostic efficacy, with sensitivity increased by 10.0% ($P < 0.001$) and specificity decreasing slightly from 98.8 to 98.2% (Fig. 4, B and C). In addition, the radiologists' reading time per case was reduced by 30.7% ($P < 0.001$) with RADAR assistance. Stratified analysis showed that RADAR assistance significantly improved performance across all reader groups. In top-tier hospitals, the performance of RADAR-assisted junior radiologists (G2) was comparable to that of unassisted senior radiologists (G1), with a 3.4%-higher sensitivity ($P = 0.24$; Fig. 4D). In other hospitals, junior radiologists (G4) improved their sensitivity by 10.1% ($P < 0.001$) when assisted by RADAR, reaching a level comparable to unassisted senior radiologists in the same tier (G3) and surpassing them by 7.0% ($P = 0.12$).

Cross-tier improvements were also observed in highly challenging or clinically critical contexts, including malignancies, emergency scenarios, and hollow-organ diseases (Fig. 4, E to G). For the detection of malignancies, RADAR assistance boosted reader sensitivity by 7.3% ($P < 0.001$), reducing missed diagnoses for these critical findings. Significant sensitivity gains were also recorded for diseases in emergency scenarios (+8.5%, $P < 0.001$) and hollow-organ diseases (+9.4%, $P < 0.001$), demonstrating RADAR's robust utility across diverse and demanding clinical tasks. Figure 4H illustrates representative cases in which unassisted readers missed critical findings; but with RADAR assistance, detection rates improved markedly and misdiagnoses were reduced. These gains were particularly pronounced for junior radiologists, suggesting that RADAR can help bridge gaps in diagnostic performance arising from differences in experience.

Performance improvements with prompt ensemble and fine-tuning strategies

To further explore potential performance improvements, we investigated two additional strategies: prompt ensemble (supplementary materials and methods "Data-driven prompt ensemble") during inference and supervised fine-tuning (Materials and methods "Fine-tuning by findings prompt alignment"). First, we introduced a data-driven prompt ensemble strategy that aggregates multiple prompts extracted from clinical reports to construct more robust textual representations. This approach achieves an AUC of 0.928 (95% CI, 0.926 to 0.929) on the internal test set, representing an improvement of 0.015 over the rule-based prompt ($P < 0.001$). Second, we fine-tuned RADAR using the proposed findings prompt alignment approach and compared its performance before and after fine-tuning (fig. S7). In the internal test cohort, the AUC significantly increased from 0.913 (95% CI, 0.911 to 0.915) to 0.940 (95% CI, 0.937 to 0.944) ($P < 0.001$). In the external test cohort, the AUC also significantly increased from 0.895 (95% CI, 0.889 to 0.900) to 0.927 (95% CI, 0.925 to 0.928) ($P < 0.001$). These results suggested that prompt ensemble and fine-tuning strategies can further enhance the performance of RADAR.

Discussion

In this study, we developed RADAR, a vision-language generalist model that fills the critical gap left by existing specialist AI models, which lack comprehensive understanding and diagnostic capability for abdominal CT scans (*37*, *38*). Compared with chest x-ray or chest CT, diagnosing abdominal diseases is considerably more difficult owing to the greater diversity of organs and disease types, substantial variations in organ morphology and spatial arrangement, and the often low contrast between pathological and healthy tissues. However, RADAR can deliver accurate diagnoses across dozens of anatomies and more than 100 clinically relevant imaging findings. This capability was enabled by three technical innovations. First, we designed a fine-grained vision-language learning framework to overcome cross-modal alignment bias caused by sparse diagnostic signals in abdominal CT. This approach requires no manual labeling and enables precise image-text association, facilitating prompt-based diagnostic inference. Second, we constructed an anatomical perception module that allows RADAR to automatically locate and segment anatomical structures during end-to-end training without external segmentation masks, further enhancing anatomical understanding. Third, we introduced an adaptive contrastive modeling strategy that leverages diagnostically relevant semantic similarities across patients to calibrate contrastive learning, ensuring accurate interpatient alignment of image and text representations.

RADAR demonstrated strong generalization, performing robustly across external centers, distinct clinical scenarios, and different imaging

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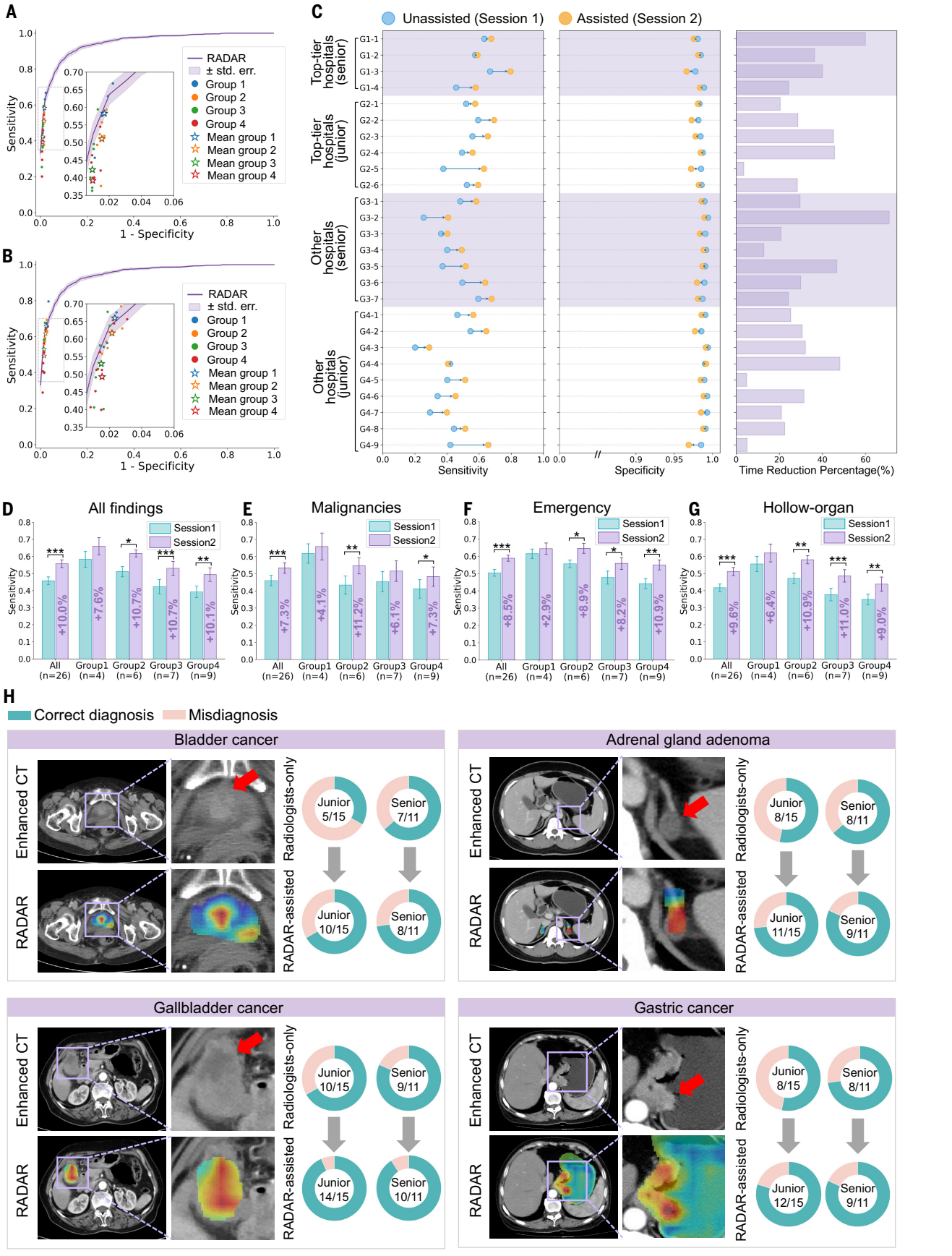


Fig. 4. Reader studies. (**A**) Comparison between RADAR and 26 readers with different levels of expertise and institutional tiers. (**B**) Performance in disease diagnosis of the same set of readers with the assistance of RADAR. (**C**) Variations in sensitivity, specificity, and the proportion of reading time reduction for different reader groups across

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two sessions. (**D** to **G**) Subgroup analysis of average sensitivity for the four reader groups, comparing performance in the unassisted versus the RADAR-assisted session across different findings categories: all findings (D), malignant diseases (E), emergency scenarios (F), and hollow-organ diseases (G). (**H**) Representative cases of diseases that were initially missed by readers but detected in the second session under the guidance of RADAR's attention maps. Error bars indicate the standard error. Asterisks denote significance: * $P < 0.05$, ** $P < 0.01$, and *** $P < 0.001$.

phases. It also generalized effectively to a cross-population cohort, achieving state-of-the-art performance on the Merlin benchmark without fine-tuning. This capacity stemmed from two key factors. First, RAD-CT, a large-scale, high-quality, and full-phase corpus from a top-tier hospital, provided the data quality and diversity that underpinned robust generalization. Second, RADAR's fine-grained framework enabled precise alignment between CT images and corresponding reports, enhancing its ability to accurately perceive and localize abnormalities, thereby reinforcing generalizability.

RADAR is scalable for both development and deployment. For development, it learned directly from raw clinical reports without manual annotation, and its fine-grained vision-language framework enabled low-cost, large-scale data expansion that improved model performance. RADAR also exhibited an unsaturated scaling behavior, suggesting potential for further improvement with more data. For deployment, it eliminates task-specific fine-tuning, allowing prompt-based queries to be applied directly to new diagnostic targets. This enables rapid adaptation to evolving clinical needs and heterogeneous case mixes while maintaining expert-level accuracy, facilitating integration into diverse health care settings. These advantages enable RADAR to achieve expert-level diagnosis with both accuracy and breadth.

The interpretability of medical AI is a long-standing focus (*39*, *40*). RADAR provides interpretable attention maps that localize relevant anatomical structures and lesions, enabling radiologists to see the regions driving each diagnostic suggestion (fig. S4). Furthermore, RADAR is able to attend to disease-specific imaging features rather than simply highlighting all abnormal regions (fig. S5). To assess the added value of such interpretability, we provided attention maps in reader studies and found that interpretable multidisease assistance reduced misdiagnoses, particularly among junior radiologists. This transparency supports informed, confident decision-making and fosters trust in human-AI collaboration, transforming RADAR from an opaque black box into a reliable assistant.

Access to high-quality imaging diagnostics remains globally unequal (*41*, *42*). Radiology services are concentrated in urban and academic centers, leaving many regions without timely and accurate interpretation (*43*). A shortage of skilled radiologists, especially in developing countries, severely limits access to accurate contrast-enhanced abdominal CT diagnoses, representing a major unmet medical need (*44*–*46*). RADAR boosts radiologists' sensitivity, enabling junior radiologists to match or exceed senior radiologists' performance. It can instantly improve diagnosis in resource-limited regions, aid training of medical students and junior radiologists. Additionally, its strong performance on non-contrast phase adds value in emergency, battlefield, and remote settings, particularly for the differential diagnosis of acute abdomen that is frequently encountered in the emergency department.

Supervised learning dominates medical imaging AI (*47*), with single-disease models achieving high AUCs, for example, various cancers such as colorectal (*48*), pancreatic (*49*, *50*), gastric (*51*), and hepatocellular carcinoma (*52*, *53*). These results come from tightly controlled settings with task-specific architectures and homogeneous data. RADAR achieved comparable AUCs on external pathology-confirmed cohorts (0.891 to 0.984), challenging the assumption that generalist models are disadvantaged. Generalist models may therefore better integrate tasks and modalities while improving scalability and reducing maintenance costs. Supervised models also show two key limitations in real-world diagnosis: instability for long-tail, data-scarce diseases and poorer external generalization due to reliance on balanced labeled data (*54*). By contrast, language-driven learning captures richer disease attributes (for example, location, morphology, and texture) beyond binary labels, supporting more scalable and robust diagnostic systems.

The limitations of this study include, first, the absence of a pathological gold standard, which remains a common challenge for generalist models (*55*). However, we performed targeted pathological validation on a subset of critical findings. This demonstrates that although this constraint affects the development phase, it does not compromise RADAR's real-world clinical utility. Secondly, the cross-population validation is not yet comprehensive. Broader testing of RADAR across more diverse populations and clinical settings is warranted.

Our work offers a dual advance for the field of AI in health care. Technically, RADAR provides a blueprint for developing future generalist AI assistants. Clinically, our ready-to-use RADAR solution represents an advance in medical practice with the potential to reshape the future of diagnosis.

Materials and methods

Ethical approval

The study was approved by the ethical committees of The First Affiliated Hospital, Zhejiang University School of Medicine (FAHZU; approval No. 2025-0992) and all the external centers participated in this study. Participant consent was waived due to the retrospective nature of this study. All data were deidentified prior to model training, testing, and reader studies.

Curation of the RAD-CT dataset

We constructed RAD-CT, a large-scale training dataset that pairs multiphase abdominal contrast-enhanced CT images with their radiological reports. The dataset was systematically compiled from a large number of abdominal contrast-enhanced CT examinations (excluding emergency cases) performed at FAHZU from January 2010 to December 2023. These data were then split into a training set and a validation set for hyperparameter tuning. The training set comprises 1,497,673 volume-wise image-text pairs, which were derived from 424,911 examinations. Although some patients may have multiple examinations over the years, we use the examination count as our primary metric. This is because the large time gaps between a patient's visits can lead to substantial differences in their imaging findings, making it more relevant to treat each examination as an independent event. A detailed flowchart of the dataset compilation process is provided in fig. S8. A core innovation of our methodology was the refinement of training data from the broad examination level to a more granular anatomy level. To accomplish this, we designed a pipeline to establish a precise alignment between specific anatomical regions within the CT images and their descriptive text in the radiological reports. This process was executed in two key steps: (i) CT scan anatomy parsing and (ii) report decomposition by anatomical region. The validation set, containing 3080 examinations, is distinct from the training set by patient. During evaluation, only findings with at least five positive examinations in the validation set were included.

Anatomy parsing: For each CT scan, we first performed automated segmentation using TotalSegmentator (*56*), which identified 104 distinct anatomical structures. Recognizing that radiological reports often describe broader anatomical regions rather than these highly granular structures, we consolidated the 104 substructures into 36 primary anatomies (table S10).

Report decomposition: Each raw radiological report was decomposed into anatomy-specific text snippets by using the Qwen language model

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(*57*). We independently processed the findings and impression sections to extract descriptions for each anatomy, which were then integrated. To handle cases of incomplete information, we applied the following rules: If an anatomy was mentioned in only one section, a "null" placeholder was used for the absent part before concatenation. To ensure a complete set of image-text pairs, anatomies not mentioned in a report were automatically assigned a "no significant abnormalities" label, consistent with standard clinical practice.

Design and training of RADAR

Network architecture: RADAR consists of a vision branch and a text branch (fig. S9), which jointly learn fine-grained alignment between anatomical structures in CT volumes and their corresponding descriptions in radiology reports. Each training sample contains a CT volume paired with its radiology report. During training, both the image and the report are decomposed into multiple anatomy-specific components so that the model learns representations at the anatomical level.

The vision branch employs a 3D U-Net encoder to extract dense volumetric feature maps from the CT scan. A segmentation decoder serves as an anatomical perception module, enabling spatial localization of multiple anatomical structures within the volume. Both the encoder and decoder are automatically configured using the nnU-Net framework based on the voxel resolution and image size of the training dataset. For each anatomy, the predicted segmentation mask is used as a spatial reference to identify the corresponding foreground region in the encoder feature maps. Feature vectors from these regions are collected to form anatomy-specific visual representations. To obtain a compact representation for each anatomy, RADAR uses a learnable query token that aggregates spatial features through a cross-attention mechanism. The resulting representation serves as the visual embedding for that anatomical structure.

The text branch processes anatomy-specific report fragments using a BERT-base encoder (*58*) (12 Transformer layers with 768 hidden dimensions and 12 attention heads). The contextualized representation of the CLS token is used as the textual embedding corresponding to each anatomical structure. More architectural details are provided in supplementary materials and methods "Network architecture."

Anatomical perception by segmentation: To enable anatomy-aware visual feature extraction, the vision branch was initially pretrained on the TotalSegmentator dataset (*56*), which provides fine-grained voxel-level annotations for multiple organ subregions. The anatomical perception module is a segmentation decoder appended to the vision encoder to localize anatomical structures. It was first pretrained on the TotalSegmentator v1 training dataset following the nnU-Net protocol, with the original 104 substructure masks aggregated into 36 main anatomical categories (table S10). For instance, all pulmonary lobes were merged into a single "lung" class. Anatomies not visible in abdominal CT scans (for example, brain, face) or those with extremely low prevalence (for example, gluteus, iliopsoas) were excluded. This yielded a final set of 18 target anatomies, which were used consistently for both training and evaluation in RADAR. The segmentation decoder was trained to predict masks for all target anatomies, thereby learning spatially localized representations. In addition, we applied the public TotalSegmentator model to the entire training set to generate organ masks, which served as pseudolabels for later training stages. These pseudolabels were used to further optimize the segmentation objectives, thereby preserving RADAR's anatomical perception capability. A combination of Dice loss and cross-entropy loss was computed between the predicted masks and the pseudolabels to refine the segmentation targets.

Anatomical image-text alignment: To mitigate alignment bias caused by heterogeneous anatomical complexity, RADAR performs fine-grained image-text alignment at the anatomical level using contrastive learning. During training, CT volumes are randomly cropped (patch size 96 by 256 by 384), and only anatomies that remain structurally complete within the cropped region are included in the alignment process. For each anatomical structure, RADAR aligns the visual embedding derived from the CT scan with the corresponding textual embedding extracted from the radiology report. The model learns to associate visual and textual representations of the same anatomy while distinguishing them from representations of other samples in the mini-batch. This anatomy-level alignment allows the model to capture localized relationships between imaging features and descriptive clinical language. The alignment objective is jointly optimized with the segmentation objective during training. Additional mathematical formulations of the loss function are provided in supplementary materials and methods "Anatomical image-text alignment."

Adaptive contrastive modeling: To mitigate false negatives arising from semantic similarities among samples from different patients, RADAR employs an adaptive contrastive learning strategy. In addition to default image-text identity matches, the model incorporates additional supervision based on anatomical status and semantic similarity. For anatomies labeled as normal, representations from different patients are treated as additional positive pairs to encourage feature proximity. For abnormal cases, similarities between pathological descriptions are estimated using a momentum-updated text encoder, enabling semantically related abnormalities to be partially aligned through soft supervision. This adaptive labeling mechanism is incorporated into the contrastive objective to improve the robustness of anatomy-level image-text alignment. Mathematical details are provided in supplementary materials and methods "Adaptive contrastive modeling."

Implementation details: All CT volumes were resized to 1 mm by 1 mm by 5 mm, with Hounsfield Unit (HU) values clipped to the range [−300, 400] and subsequently normalized to [0, 1]. During training, volumes were randomly cropped into patches of size 96 by 256 by384 voxels, which served as the model input. Only complete organs within each patch were considered, and any partially included organs due to cropping were excluded from training. RADAR was trained for 30 epochs using the Adam optimizer, with an initial linear warm-up of the learning rate to 1×10^{-4} in the first epoch, followed by a cosine decay to 1×10^{-6}. The training was performed on 24 NVIDIA H20 GPUs with a total batch size of 48. For the supervised learning baseline, training labels were automatically extracted from the corresponding reports using our fine-tuned Qwen2.5-7B model.

Model inference

To improve inference efficiency, RADAR adopts a sliding-window strategy with a patch size of 96 by 256 by 384 and a stride equal to half of the window dimensions. For each patch, the segmentation decoder generates anatomical masks and evaluates the completeness of each anatomy to determine whether the organ extends beyond the patch. If an organ is found to extend beyond the current window, it is deemed incomplete and excluded from further processing in that window. If none of the sliding windows can capture a complete organ, a center-crop operation is applied to extract the organ separately. For each complete anatomy, organ localization is guided by its mask, and the corresponding anatomical features are extracted from the visual representation space.

RADAR used rule-based prompt querying to perform diagnostic decision-making. For each finding, both positive and negative textual prompts were constructed. Rule-based prompts were first designed according to radiology reporting conventions. For imaging findings, commonly used descriptive phrases from reports were directly used as positive prompts (for example "bladder wall irregularity"). For disease entities, positive prompts were generated by combining typical

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descriptive phrases with the disease name (for example "pancreatic swelling, peripancreatic exudation" and "pancreatitis"). Negative prompts followed standard reporting patterns, including phrases such as "no {finding}" and "no evidence of {finding}".

To improve robustness to linguistic variation, we further introduced a data-driven prompt ensemble strategy (supplementary materials and methods "Data-driven prompt ensemble"). For each finding, report sentences from the training set were used to construct sets of positive and negative prompts. Descriptions associated with positive cases formed the positive prompt set, whereas statements indicating the absence of the finding or normal anatomy formed the negative prompt set. Each prompt was encoded by the text encoder, and embeddings within each set were aggregated by mean pooling to obtain ensemble representations for the positive and negative prompts of that finding.

At inference, cosine similarity was computed between the anatomical visual embedding and the positive and negative prompt embeddings. The probability of the finding was obtained through a softmax over the similarity scores. When the same anatomy was captured across multiple sliding windows, predictions were averaged to improve stability. For multiphase CT examinations, predictions from the nonenhanced, arterial, and venous phases were further ensembled during inference. Specifically, the maximum prediction score across the three phases was used as the final ensemble score.

Benchmark and evaluation

Internal testing cohort: For internal validation, we collected an internal testing cohort of 39,160 examinations from FAHZU, encompassing all abdominal contrast-enhanced CT examinations performed from January to June 2024. This cohort was specifically constructed to ensure a clear temporal separation from the RAD-CT training data.

Emergency testing cohort: For emergency validation, a test cohort of 27,267 examinations was assembled from FAHZU. The examinations, collected prior to January 2024, are entirely distinct from the internal training and testing datasets. This cohort includes 17 common acute abdominal conditions, determined based on their prevalence (table S1).

External testing cohorts: For external validation, we included eight independent cohorts, encompassing a total of 24,239 examinations, to assess the generalizability and robustness of RADAR. The participating centers included: Anji County People's Hospital (Center A, 4808 examinations, 118 findings), People's Hospital of Beilun District (Center B, 2200 examinations, 86 findings), Haining People's Hospital (Center C, 4373 examinations, 131 findings), People's Hospital of Jingning She Autonomous County (Center D, 783 examinations, 68 findings), Jixi County People's Hospital (Center E, 640 examinations, 71 findings), The First People's Hospital of Yuhang District (Center F, 1436 examinations, 115 findings), Shengzhou People's Hospital (Center G, 8040 examinations, 136 findings), The First Division Hospital of Xinjiang Production and Construction Corps (Center H, 1959 examinations, 93 findings).

Pathology-confirmed testing cohorts: Although our model was trained on report-based diagnoses, we further validated its generalizability using pathology-confirmed gold standard labels. For this purpose, we enrolled a total of 4333 patients from two independent cohorts at The Affiliated Hospital of Jiaxing University (pathology-confirmed center 1, totally 2001 examinations) and Shengzhou People's Hospital (pathology-confirmed center 2, totally 2332 examinations), which included cases of colorectal, pancreatic, and gastric cancers, as well as hepatocellular carcinoma.

Findings inclusion: The inclusion of findings for extensive evaluation purposes was conducted through a structured, three-stage process to ensure comprehensive clinical relevance and practical utility. (i) Initial clinical curation: Two experienced radiologists systematically extracted and compiled all clinically relevant diagnostic findings—including diseases and imaging signs—that appeared in a large corpus of real-world abdominal CT reports from the Picture Archiving and Communication System of FAHZU. This bottom-up approach ensured that the finding list reflected the actual terminologies, frequency, and diagnostic focus of routine radiological practice. This initial list broadly covered a wide spectrum of abdominal abnormalities, including hepatic, biliary, pancreatic, gastrointestinal, renal, vascular, splenic, and urinary tract lesions commonly encountered in abdominal imaging. (ii) Frequency-based pruning: To refine the candidate list, we performed a large-scale statistical analysis of more than 200 preliminary findings using massive abdominal contrast-enhanced CT radiology reports from our primary institution. The frequency of each finding's occurrence was quantified based on its mention in the clinical reports. Findings with low prevalence, such as rare conditions (for example hepatic angiosarcoma), were excluded to ensure sufficient representation and clinical utility in downstream modeling. Owing to the limited segmentation capability of TotalSegmentator for certain anatomies, we excluded diseases involving organs without an available segmentation model, such as the peritoneum, ureter, ovary, and uterus, from this step. (iii) Expert-driven refinement: Two chief radiologists reviewed the remaining findings to optimize diagnostic value and label interpretability. Terms lacking pathological specificity or clinical utility were removed. This included vague descriptors often related to scan quality or protocol variations (for example "suboptimal bladder filling", "poor pancreatic enhancement"), signs with low diagnostic relevance (for example "mesenteric fat stranding"). Through this systematic, multitiered selection process, we finalized a list of 146 abdominal findings that are clinically meaningful, reliably detectable on contrast-enhanced CT, and well-suited for practical application and interpretability in AI-driven diagnostic systems. These curated findings provide a foundation for evaluating diagnostic algorithms in real-world abdominal imaging scenarios.

Emergency findings inclusion: To determine diseases in emergency scenarios, a chief radiologist first identified all clinically relevant conditions in the emergency setting. These diseases were then mapped to our 146 findings, and diseases with fewer than 10 positive cases in the emergency testing cohorts were excluded for stability and reliability of evaluation. Consequently, 17 representative emergency diseases were ultimately selected to serve as benchmarks for evaluating the model performance in emergency clinical applications.

Extraction of findings: We introduced a semiautomated approach for extracting findings from radiology reports, comprising three key steps: (i) Instruction fine-tuning of the Qwen2.5-7B model. The objective is to determine, from radiology reports, the presence or absence of specific findings. To construct the instruction dataset, 3080 reports from validation set were manually annotated by an experienced radiologist, which were then split into training (80%) and validation (20%) subsets. For simplification, the 146 findings were divided into six groups, each containing 20 to 30 findings, with a separate Qwen2.5-7B model fine-tuned for each group. For instance, in the case of a Qwen model tasked with classifying 25 findings, we devised the following instruction: "From the description and conclusion sections of the given CT report, determine whether specific findings are mentioned. You need to identify the following 25 findings, such as whether the kidney mentions renal pelvis dilatation, whether the kidney mentions hydronephrosis, ..., and whether the duodenum mentions a tumor. Please return the results strictly in the following json format: {anatomy_finding: yes/no, anatomy_finding: yes/no, ..., anatomy_finding: yes/no"}. We integrated the description and conclusion sections of the CT report into one prompt: "Report description: {Description}, report conclusion: {Conclusion}". To further enhance the model's ability to recognize findings, we supplied relevant medical anatomical knowledge, such as

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"The colon includes the cecum, colon proper, rectum, and anal canal. The cecum contains the appendix, and information related to the appendix should be associated with the colon. The small bowel comprises jejunum and ileum, while the splenic vein is part of the portal venous system." Consequently, the input format for the model is "Instruction + Prompt + Knowledge", and the output format is "{anatomy_finding: yes/no, anatomy_finding: yes/no, ..., anatomy_finding: yes/no}". The accuracy of extracted 146 findings obtained by this method was provided in table S11. (ii) Prompt-based extraction using Qwen2.5-Max API. Through prompt engineering, the Qwen2.5-Max model was queried to automatically detect explicit mentions of specific findings within the reports. The prompt was designed as follows: "Report description: {Description}, report conclusion: {Conclusion}. Based on this CT report, indicate whether {finding} is mentioned. Please respond with: yes or no". Additionally, we provided two supplementary inputs: (a) relevant medical anatomical knowledge, as referenced in "Instruction fine-tuning the Qwen2.5-7B model as findings extractor" section; (b) guidance for Qwen to focus on specific sections of the report depending on the finding type. For instance, if the finding was classified as a disease, Qwen was directed to prioritize the conclusion section; if the finding was a sign, emphasis was placed on the description section. The accuracy of the extracted 146 findings obtained by this method is provided in table S12. (iii) Expert adjudication of final labels. The outputs from steps (i) and (ii) were compared. Cases with inconsistent predictions were manually reviewed by radiologists to establish the final label. For cases with consistent predictions, a random 10% sample was audited to ensure quality control. Through this three-step process, we assigned labels to 146 findings in the internal and external testing cohorts. This systematic approach not only ensures high-accuracy finding labels but also substantially reduces manual annotation workload.

Evaluation metrics: Each diagnosis task was treated as a binary classification problem, evaluated using the AUC score. We computed the AUC for each disease category and then averaged these values to obtain the general AUC across multiple tasks. For cases in which an organ was in a postoperative state, diagnostic tasks related to that organ were excluded from the evaluation.

Cross-organ comorbidity analysis: To evaluate the model's capacity to handle clinical comorbidities and ensure that organ-specific contrastive training does not compromise diagnostic performance for co-occurring conditions, we analyzed five PVH-related findings: (liver) cirrhosis, (portal vein) hypertension, (esophagus) varices, (stomach) gastric fundal varices, and (spleen) splenomegaly. First, using a cohort of 258 patients confirmed to have all five conditions, we assessed the recall performance of the model across varying clinical sensitivity requirements by calculating the number of correctly identified comorbidities under three distinct specificity thresholds (90, 95, and 99%), as illustrated in fig. S1A. Furthermore, we evaluated the discriminative power of diagnostic scores via pairwise analysis (fig. S1B). By examining the joint distribution of predictions for co-occurring comorbidities, we assessed the model's ability to distinguish "neither-disease" and "both-disease" groups. An effective model should demonstrate clear separation: "neither-disease" samples cluster near the origin, while "both-disease" samples aggregate toward the upper-right, reflecting simultaneous high-risk predictions. Finally, to quantify diagnostic consistency (fig. S1C), we performed 5000 bootstrapping iterations where we randomly sampled pairs consisting of one "neither-disease" and one "both-disease" case. By measuring the score shift between these paired samples, we computed a consistency score reflecting the model's propensity to concurrently elevate (+1) or decrease (−1) risk scores for associated conditions, thereby validating the model's ability to generate synchronized risk profiles that reflect the clinical synergy of these comorbidities.

Model interpretation

We applied Grad-CAM to identify the image regions attended to by the vision-language model during classification. Specifically, the cosine similarity between the positive prompt and the anatomy features served as the prediction score. Gradients with respect to this score were computed via backpropagation and used to generate an attention map, in which higher values correspond to regions more influential to the model's decision.

Reader studies

To assess the clinical utility of RADAR as an assistive tool for radiologists, we conducted two sequential reader studies. The first study evaluated the diagnostic performance of radiologists with varying levels of experience from different institutions and compared it with that of RADAR. The second study assessed whether RADAR could improve the accuracy of diagnostic reports. A total of 26 radiologists from 14 institutions participated, stratified into four groups: senior radiologists from top-tier hospitals ($n = 4$), junior radiologists from top-tier hospitals ($n = 6$), senior radiologists from other hospitals (n = 7), and junior radiologists from other hospitals ($n = 9$). Seniority was defined as >10 years of clinical experience; junior level was defined as ≤10 years of experience. Institutional stratification followed the Chinese Hospital Grading Standards administered by the National Health Commission (*59*). Top-tier hospitals were defined as Grade III-A (the highest grade) institutions, and other hospitals as Grade III-B and Grade II-A institutions. This grouping was used for reader stratification and for all subgroup analyses.

A total of 300 cases were sampled from the internal testing cohort, selected to cover a broad spectrum of diseases and complex presentations. The final set comprised 61 distinct imaging findings (table S1). All studies were performed on a purpose-built web-based NIfTI viewing platform equipped with essential tools, including HU windowing, zoom, and multiplanar reconstruction (fig. S10). Multiphase abdominal CT scans, including nonenhanced, arterial, and venous phases, were provided concurrently for each case. Before the study, all readers underwent standardized training and were instructed to review each case carefully and write a structured diagnostic report containing both description and conclusion sections. No time limits were displayed, but the platform recorded the reading time for each case.

In the first session, readers interpreted cases independently without AI assistance. After a washout period of at least 1 month to minimize memory bias, the same readers reinterpreted the cases in the second session with RADAR assistance, which provided predicted positive findings and corresponding attention maps. Ground truth labels for each finding were extracted from original clinical reports and manually reviewed for accuracy. Predicted findings from each reader's report were extracted using the same process.

Ablation study

To evaluate the contribution of each component in RADAR, we conducted extensive ablation experiments on the internal validation set, focusing on two main aspects: architectural design and the anatomy-level contrastive learning paradigm. For architectural design, we evaluated alternative visual backbone architectures and assessed the role of the anatomical perception module by comparing models trained with and without this component. For the alignment framework, we examined several design choices within the anatomy-level contrastive learning paradigm, including the use of anatomy-level versus whole image–text alignment, the adaptive contrastive loss versus the standard InfoNCE formulation, different strategies for aggregating anatomical features, and the momentum-based similarity weighting mechanism. More details are provided in supplementary materials and methods "Ablation study."

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RADAR+: Extending diagnostic scopes

To extend RADAR beyond organ-specific tasks, we introduced a global image-text alignment module in RADAR+, enabling the model to associate whole-volume CT representations with complete radiology reports. This design is particularly useful for diagnostic scenarios in which the relevant pathology is not confined to a predefined anatomical structure or may involve multiple distributed regions. In the vision branch, a learnable global query token aggregates spatial features from the entire CT feature map through a cross-attention mechanism, producing a global visual representation of the scan. In the text branch, the full radiology report is encoded by the text encoder to obtain a corresponding global textual representation. These global visual and textual embeddings are then aligned using a contrastive learning objective that encourages representations from the same patient to be closer than those from different patients. Training of RADAR+ adopts an interleaved optimization strategy. Within each training epoch, mini-batches are alternately used for anatomy-level alignment and global alignment. This strategy preserves the anatomy-specific discriminative capability learned through organ-level supervision while simultaneously enabling the model to capture holistic diagnostic information at the whole-volume level. More details of the global alignment module are provided in supplementary materials and methods "RADAR+."

Fine-tuning by findings prompt alignment

To further enhance the model's ability to recognize specific clinical findings, we introduced a findings prompt alignment strategy and performed additional fine-tuning. In this approach, clinical findings associated with each anatomy are automatically extracted from radiology report fragments using the Qwen LLM. For each finding, paired textual prompts are constructed to represent its presence and absence. These prompts are encoded by the text encoder and aligned with the corresponding anatomy-level visual representations produced by the vision branch. During fine-tuning, the model is optimized to increase the similarity between anatomical visual features and prompts describing true findings while decreasing similarity with prompts describing absent findings. This procedure strengthens the correspondence between localized visual representations and explicit diagnostic descriptions. More details are provided in supplementary materials and methods "Findings prompt alignment."

Details of compared methods

We compare RADAR against several representative vision-language and supervised learning baselines.

CT-CLIP: CT-CLIP is a 3D adaptation of CLIP, originally designed for natural image-text pairs. To accommodate volumetric CT data, it employs the Vision Transformer as its 3D vision encoder and CXR-BERT as the text encoder. CT-CLIP performs whole-volume contrastive learning by directly aligning the entire CT volume with the corresponding patient-level radiology report, which contains multiorgan textual descriptions. To ensure compatibility with Chinese radiology reports in RAD-CT, we reimplemented CT-CLIP using the same pretrained Chinese BERT as RADAR's text encoder. Moreover, we reimplemented CT-CLIP with the same prompt strategy as RADAR during inference.

Merlin: Merlin is another 3D CLIP variant that shares a similar architecture with CT-CLIP but uses a pretrained 3D ResNet-152 as the vision encoder and Longformer as the text encoder. Similar to CT-CLIP, Merlin performs whole-volume contrastive learning by directly aligning the entire CT volume with the corresponding patient-level radiology report. However, it jointly optimizes contrastive learning and a phenotype classification objective during training. The reimplemented Merlin also adopts the same Chinese BERT text encoder and prompt strategy as RADAR. Since RAD-CT does not include phenotype annotations, the phenotype classification loss was omitted during retraining.

fVLM: fVLM is a fine-grained visual-language model previously proposed by our team that performs anatomy-level contrastive learning. Unlike whole-volume alignment approaches, fVLM first obtains anatomical masks using TotalSegmentator and leverages the Qwen language model to parse the patient-level report into anatomy-specific textual descriptions. It then aligns visual and textual representations at the anatomical level. To mitigate false negatives inherent in anatomy-level contrastive learning, fVLM introduces a dual false-negative reduction strategy. While fVLM shares the same Chinese BERT text encoder as RADAR, it uses a Vision Transformer as its visual backbone, rather than RADAR's CNN-based encoder. Crucially, fVLM relies on external segmentation from TotalSegmentator and does not learn anatomical localization end-to-end. During inference, it also adopts RADAR's sliding-window strategy and prompt strategy.

BiomedCLIP: BiomedCLIP is a domain-agnostic foundation model pretrained on large-scale 2D biomedical image-text pairs via contrastive vision-language learning. We evaluated BiomedCLIP using a prompt-based contrastive inference strategy. Specifically, for each finding, the model calculated similarity scores for each 2D slice by comparing its image embedding against a positive prompt ("This is an image of <finding>") and a negative prompt ("This is an image showing no <finding>"). The final scan-level prediction was then obtained by averaging these similarity scores across all constituent slices of the 3D volume, a procedure consistent with Merlin.

MedGemma-1.5-4B-it: MedGemma is a medical large multimodal model built upon the Gemma 3 architecture and further optimized for advanced performance in medical text and 2D image comprehension. MedGemma-1.5-4B-it is an updated iteration that enhances support for 3D volumetric data by partitioning volumes into a composite sequence of 2D slices, which are processed as a unified multimodal input. To assess its diagnostic capabilities, we adopted Vision Question Answering (VQA) strategy. Specifically, for each 3D CT, the model was prompted with a question: "Does this CT slice show any evidence of <finding>. Answer with only 'yes' or 'no'." Diagnostic scores were derived by calculating the softmax-normalized probability of the positive ("yes") versus negative ("no") tokens. The resulting positive token scores were then used to determine the AUC for performance benchmarking.

Lingshu-7B: Based on the Qwen2.5-VL architecture, Lingshu-7B utilizes a multistage training paradigm and a rigorous data curation pipeline to align general-domain AI capabilities with clinical demands, such as report generation and multimodal question answering. Notably, the model is uniquely integrated with a reinforcement learning paradigm and verifiable rewards to enhance complex clinical reasoning. Its diagnostic capabilities were assessed using a 2D-based VQA strategy, employing the same prompt as MedGemma-1.5-4B-it for each slice. Diagnostic scores were derived from the softmax-normalized probability of the "yes" versus "no" tokens. Following the protocol in Merlin, these slice-level probabilities were averaged to yield the final scan-level prediction for the 3D volume.

Supervised learning model ViT: We implemented a ViT-base architecture, initialized with MAE ImageNet-1K pretrained weights, as the vision encoder. CT volumes are split into 16 by 16 by 32 patch tokens. For each finding task, we appended a newly introduced CLS token, concatenated with the image tokens, as input to the Transformer. Each finding has an independent classification head (two-layer MLP), with the CLS token as input, predicting the diagnosis probability. The model was trained end-to-end using a multilabel binary cross-entropy loss. The supervised baseline used the same data preprocessing pipeline as RADAR, including the same CT intensity normalization protocol, input patch size (96 by 256 by 384 voxels), and cropping strategy. The core optimization settings were also aligned with RADAR, including

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the Adam optimizer and a batch size of 48. For supervised training, the initial learning rate was set to 1×10^{-4} with linear warm-up during the first epoch and was reduced by half if the validation loss did not decrease for three consecutive epochs. During inference, the same sliding-window strategy as RADAR was used.

Supervised learning model nnU-Net+: Based on segmentation framework of nnU-Net, we develop nnU-Net+, a multitask supervised learning model that jointly performs anatomical segmentation and finding classification. The network architecture is automatically configured by the nnU-Net toolbox, and was kept the same as the RADAR. Similar to RADAR, organ masks were obtained using TotalSegmentator, enabling an organ segmentation pretext task. For classification, anatomy-specific visual representations were extracted via the organ mask and processed using a cross-attention mechanism similar to RADAR's alignment step. Following RADAR's design principle, nnU-Net+ extracts anatomy-specific image embeddings using the predicted segmentation masks. Anatomy-wise learnable queries then attend to each anatomical embedding via cross-attention to produce the CLS token per anatomy. For each anatomy, classification heads (two-layer MLP) were added according to the number of findings associated with that anatomy. Each head takes the anatomy CLS token as input and outputs the diagnostic probability for its target finding. Multilabel binary cross-entropy loss was used for optimization. Implementation details were the same as those described above for the supervised ViT baseline.

Statistical analysis

Confidence intervals were calculated based on 1000 bootstrap replications of the corresponding data. The significance comparisons of sensitivity and specificity were conducted using permutation tests to calculate two-sided P values with 10,000 permutations. Comparisons of AUCs were conducted using DeLong's test. Data are presented as mean ± standard error or 95% CI, as appropriate. The threshold to determine significance was $P < 0.05$. Data analysis was conducted in Python using the numpy (v.1.26.4), scipy (v.1.14.1), and scikit-learn (v.1.6.0) packages.

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SUPPLEMENTARY MATERIALS

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Materials and Methods; Figs. S1 to S13; Tables S1 to S13; MDAR Reproducibility Checklist

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An expert-level generalist AI for abdominal CT diagnosis

Qi Zhang, Jianpeng Zhang, Weiwei Cao, Zilin Lu, Wanxing Chang, Haonan Ding, Cao Chen, Zhi Li, Xing Xue, Sinuo Wang, Shaoteng Zhang, Yutong Xie, Yong Xia, Qi Wu, Zhongyi Shui, Xi Li, Zhilin Zheng, Yanjie Zhou, Tony C.W. Mok, Yingda Xia, Hongkan Wang, Xianghua Ye, Tao Ma, Jie Peng, Xiaoguang Wang, Jian Ding, Yuming Gao, Huazhen Ye, Yiping Liu, Dongjie Chen, Zhaomin Ni, Jianwen Ning, Wei Zhang, Jian Liu, Chaohui Yu, Shenghong Ju, Jianfeng Zhang, Wenbo Xiao, Ling Zhang, and Tingbo Liang

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Editor's summary

Contrast-enhanced abdominal computed tomography (CT) is a common imaging modality used as part of the diagnostic evaluation for numerous conditions. It is also one of the more difficult to evaluate because of the number of anatomical structures in the abdomen and the fact that clinically important findings can present as subtle abnormalities that are easily missed. Previous attempts at developing artificial intelligence (AI) solutions relied on supervised learning and human annotations, making them very labor intensive to train. Zhang *et al*. developed a generalist vision–language model called RADAR, which demonstrated strong performance without manual annotation, even for abnormalities it had not encountered during training, and it was also effective at augmenting the diagnostic success of human radiologists. —Yevgeniya Nusinovich

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